CMA Stormwater Ditch List and Maps

(as of 2022-03-19)

Attachments depict and list currently known CMA stormwater ditches, approximate lengths, flow directions, and notes from a ditch survey conducted in March 2022. Weather and time limited information gathered. Pictures and geo-coordinates were taken for each ditch but are not included due to space; they are available. Mapped ditches are in lite fuchsia with an AssetID corresponding to same field in the spreadsheet. This information should be used in conjunction with the culvert map/list provided earlier. In most cases ditches end at a culvert with many continuing on the culverts other end. A few end at/near existing intermittent streams and are marked "STRM" in the AssetID; some have downed trees across/along them that do or may block their path and may eventually change stormwater run-off routes. These need to be monitored for needed action by the Site Manager, Board, and members.

Conditions found needing attention: All ditches are impacted by leaves to some extent. In some cases, where they end at large diameter (>24") culverts and sufficient culvert slope exists, the leaves do not create a problem and need no action unless tree limbs, logs, or rocks in the ditches or at culvert entrances prevent or divert stormwater flow. In many cases, the ditches are or start as relatively shallow depressions that leaf blowers can be used to clear, except where they meet culverts, or have limbs, logs, or bushes blocking flow. Several ditches are sufficiently deep that scraping by hand tools, backhoe, or similar equipment will be necessary. Downed trees, logs, and large rocks need to be removed to prevent clogs and disposed of or placed where stormwater flow won't be impacted or wash debris back into ditches. A few ditches appear to "die out" or end where natural channels to streambeds have filled in (ex: MtnSdRd_19). The channels should be re-established or verified as working during a heavy rain. In one case, MtnSdRd_28, seems intended to flow to ditch RcWy_02, a near 90° turn; the amount of runoff anticipated and steepness appears better served by installing a culvert under Recreation Way connecting with MtnSdRd_31; this may necessitate enlarging ditches and culverts downstream on the west side of Mountainside Rd to ditch MtnSdRd_33. Some ditches, usually adjacent culverts, need rip-rap to help control erosion or sediment traps; these need to be carefully considered as they will require their own maintenance due to sediment/debris collection that could necessitate heavy equipment use.

Recommendations:

Include in contract: 1) Ditches and culverts are of necessity intertwined; cleaning out culverts will necessarily impact the connected ditches. They may be best served by being incorporated into one contract effort. Alternately, identifying and requiring appropriate disposal of culvert clean-out is paramount. **2)** Incorporate in the contract a requirement to note needed changes, corrections, modifications of the ditch map and list. **3)** Identify proper disposal locations or a requirement to haul away leaves, limbs, logs, or rocks vice just piling up adjacent roadways or just outside or at the tops of ditches where they're unsightly and stormwater runoff or erosion can return them to the ditches. **4)** Some ditches have vegetation (numerous saplings and bushes) growing in them that hold leaves in place and or blocks needed water flow; these need to be removed.

Future program: 1) Advise residents of the need for their help in identifying stormwater run-off issue areas and establish a method and point of contact for such reports; examples include: vegetation growth, fallen trees or logs or large rock falls in ditches; large run-offs of the watershed not entering ditches during downpours. **2)** Program regularly recurring reviews of stormwater controls (ditches, culverts, streams, watershed during heavy storms by the Site Manager or other identified position to verify installed or needed stormwater infrastructure. **3)** Ensure DACC regs include stormwater runoff as a consideration in all residential and other building, road, and driveway projects for tie-in with existing natural and man-made drainage for location, sizing and inter-connection. **4)** Some ditches appear to be on private property, such as on Warbler Way, the Board needs to establish policy on ditch maintenance responsibility, i.e. homeowner or CMA.

CMA Culvert List
(as of 2022-03-19)

Asset ID	Length	Diameter	Material	Notes
BctDr_1	24	22"	Galvanized	22” actual update
BctDr_10	36	24"	Galvanized	The west side of this culvert is filling in with rock because it’s so deep and needs to have a cement concrete barrier put up around it.
BctDr_11	47	16"	Galvanized	
BctDr_12	14	14"	Galvanized	Desperate for cleaning.
BctDr_13	30	14"	Galvanized	Both ends of culvert are impacted by vegetation and hard to find.
BctDr_14	38	14"	Galvanized	
BctDr_15	27	14"	Galvanized	The southern end of the culvert is below ground about 2'; the northern end is underneath rock and we could not actually put hands on the culvert, but did find water running out of it.
BctDr_16	35	24"	Galvanized	
BctDr_2	36	40"	Galvanized	
BctDr_3	31	18"	Galvanized	
BctDr_4	30	14"	Galvanized	Has walls around both ends of pipe to support/trap sediment.
BctDr_5	35	12"	Galvanized	
BctDr_6	68	20"	Galvanized	Northern end of culvert is too close to surface causing cracks in the road.
BctDr_7	26	24"	Galvanized	Culvert is rusting badly and has wooden flume on the eastern end extending 20-30' minimum and needs repair.
BctDr_8	18	16"	Galvanized	
BctDr_9	27	16"	Galvanized	Actual diam 16”
BctLn_1	34	Unknown	Unknown	Culvert is HDPE is HDPE on one end, cement on the other, with different diameters.
BctWy_1	44	20"	Galvanized	One end of the pipe has both a galvanized and an HDPE plastic pipe, on the other end we could only locate galvanized, not the HDPE pipe.
CdCt_1	15	14"	Concrete	
CdLn_1	20	12"	Concrete	Northern side is visible concrete culvert, the southside could not be found, it appears to be totally impacted.
CdLn_2	11	14"	Concrete	
CdLn_3	18	16"	Concrete	
				The northwest end of this culvert is impacted by brambles the only way to find it is to look for the horizontal telephone pole adjacent the road, the southside is open and available.
DgWd_1	41	48"	HDPE	Downhill outfall needs work, rocks here originally but they’ve been taken away and may need to be replaced
DgWd_2	28	12"	Galvanized	
DgWd_3	30	12"	Galvanized	North end of culvert could not be located the south end of the culvert is totally impacted
DgWd_4	52	20"	Concrete	
DgWd_5	22	16"	Concrete	Abandon, appears non-functional
DgWd_6	52	18"	Concrete	West end filled with tree limbs and leaves
FrgHlw_1	16	24"	Galvanized	Location has two pipes, appears due to road erosion during a heavy rain.
FrgHlw_2	21	14"	Galvanized	This is part of a two pipe installation, the larger one is 24", this one 14"

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Asset ID	Length	Diameter	Material	Notes
FrgHlw_3	18	12"	Concrete	This is part of a two pipe installation, one is 24" galvanized, this one is 12" cement
FrgHlw_4	12	24"	Galvanized	This culvert is a galvanized, there is a smaller cement culvert about 3 feet further north.
HbtQk_1	37	12"	Galvanized	
HbtQk_2	15	24"	Galvanized	
HbtQk_3	29	12"	Galvanized	
MtnSdCt_1	16	14"	Galvanized	75% full of dirt leaves, etc.
MtnSdRd__13	54	48"	HDPE	
MtnSdRd_1	44	30"	HDPE	A cement pipe is adjacent the black HDPE on the west side, the cement pipe appears non-functional; a galvanized adjacent the black HDPE pipe on the east side, also appears non-functional.
MtnSdRd_10_O1	16	14"	Galvanized	
MtnSdRd_11_O3	17	12"	Galvanized	
MtnSdRd_12_O4	16	16"	Galvanized	
MtnSdRd_14_O5	7	18"	Galvanized	
MtnSdRd_15_Barr	44	14"	Galvanized	
MtnSdRd_16_D4	10	12"	Galvanized	This includes a 14" inch pipe going in to a 12" pipe.
MtnSdRd_17_BluE	26	14"	Galvanized	The end of the pipe on the north end could not be found.
MtnSdRd_18	20	14"	Galvanized	
MtnSdRd_19_B1	24	18"	Galvanized	
MtnSdRd_2	45	20"	Galvanized	
MtnSdRd_20	27	Unknown	Unknown	This location has yellow stakes indicating a culvert but not culvert was found.
MtnSdRd_21_A1	47	18"	HDPE	
MtnSdRd_22_B6	33	14"	Galvanized	
MtnSdRd_23	39	30"	Galvanized	
MtnSdRd_24	29	24"	Galvanized	There are two culverts at this location one on top of the other the lower one cement the upper one galvanized
MtnSdRd_25	38	12"	Galvanized	North end is completely crushed and filled with gravel.
MtnSdRd_26	42	14"	Unknown	Culvert is galvanized on one end, cement on the other, unknown connection point.
MtnSdRd_27	34	12"	Galvanized	East end of culvert is nearly full of gravel and sand.
MtnSdRd_28	31	14"	Concrete	North end appears broken 3'-4' from end, appears 3/4 full of debris.
MtnSdRd_3	43	14"	Galvanized	Pipe has a bend about 15° down in the middle of the road unknown if it's all connected, has an elbow, or is just bent.
MtnSdRd_4	28	20"	Concrete	Pipe appears to be sleeved or individual units seal together
MtnSdRd_5	39	24"	Concrete	
MtnSdRd_6	33	14"	Galvanized	Eastern end was covered and filled in by recent tree work.
MtnSdRd_7	45	24"	Galvanized	
MtnSdRd_8	50	24"	Galvanized	
MtnSdRd_9	44	30"	Galvanized	
RResRd_1	40	14"	Galvanized	
Salmon_Ln_2	49	14"	Galvanized	Bottom rusted out.
Salmon_Ln_3	35	42"	Galvanized	This is a double pipe set; actual length is 34' each.
Salmon_Ln_4	35	42"	Galvanized	This is a double pipe set; actual length is 34' each.
Salmon_Ln_5	46	30"	Galvanized	Rusting but not rusted through.
Salmon_Ln_6	47	24"	Galvanized	

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Salmon_Ln_7	36	30"	Galvanized	West End is crushed and full of gravel.
Salmon_Ln_T2	18	14"	Galvanized	
Salmon_Ln_T4	29	12"	Galvanized	
Salmon_Ln_T5	39	18"	Galvanized	
Salmon_Ln_T6	31	30"	Galvanized	Bottom rusted in spots.
South_Res_1	18	30"	Galvanized	
South_Res_2	16	14"	Galvanized	Culvert is of questionable use because of inlet side being very narrow.
Wblr_1	59	14"	Galvanized	
Wblr_2	21	14"	HDPE	
Wblr_3	33	12"	Galvanized	Grate on south end.
Wdpkr_1	42	14"	Galvanized	North end is crushed down needs to be bent back out at a minimum.
Wdpkr_2	34	14"	Galvanized	
Wdpkr_3	44	14"	HDPE	
Wdpkr_4	43	20"	HDPE	
Wdpkr_5	30	14"	Galvanized	Downhill side of this culvert is crushed.
Wdpkr_6	31	14"	HDPE	Cullvert appears to be part of the owners driveway, probably installed by them and may be their responsibility.
Wdpkr_7	18	14"	HDPE	
WhpWl_1	32	14"	Galvanized	Both ends are protected by rock/cement enclosure.
WhpWl_2	23	Unknown	Galvanized	West end mostly filled with dirt and rock; east end appears filled in by snow plow.
WhpWl_3	17	12"	Galvanized	Half full of silt and mud; west end crushed.
WntGrn_1	40	54"	Galvanized	
WntGrn_2	36	45"	Galvanized	
WntGrn_3	25	12"	Galvanized	
WntGrn_4	23	40"	Galvanized	
WntGrn_5	21	12"	Galvanized	North end appears buried under relatively new driveway of S3.
WntGrn_6	59	12"	Galvanized	
WntGrn_7	19	12"	Galvanized	

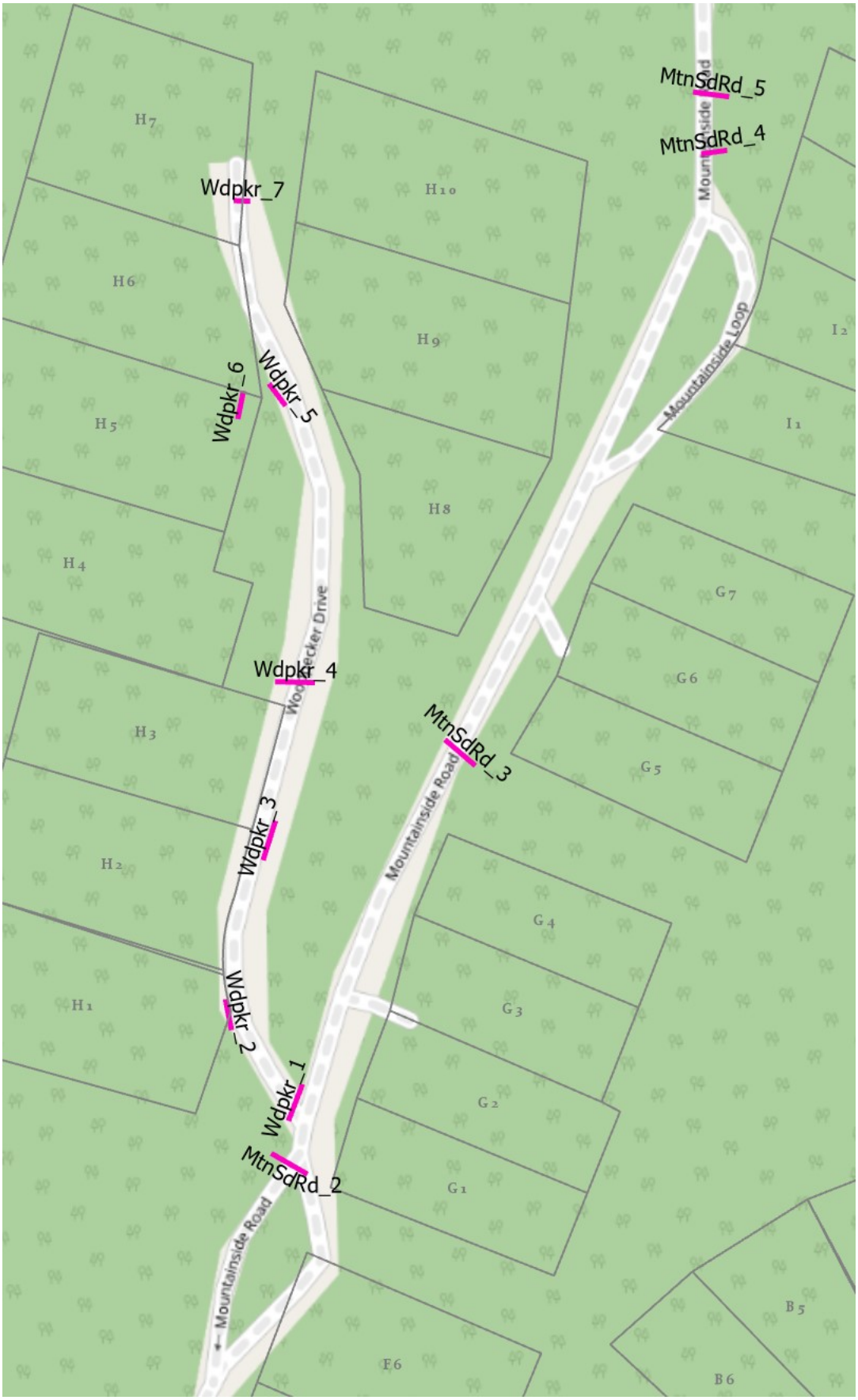
South Reservoir - Salmon Lane Culverts



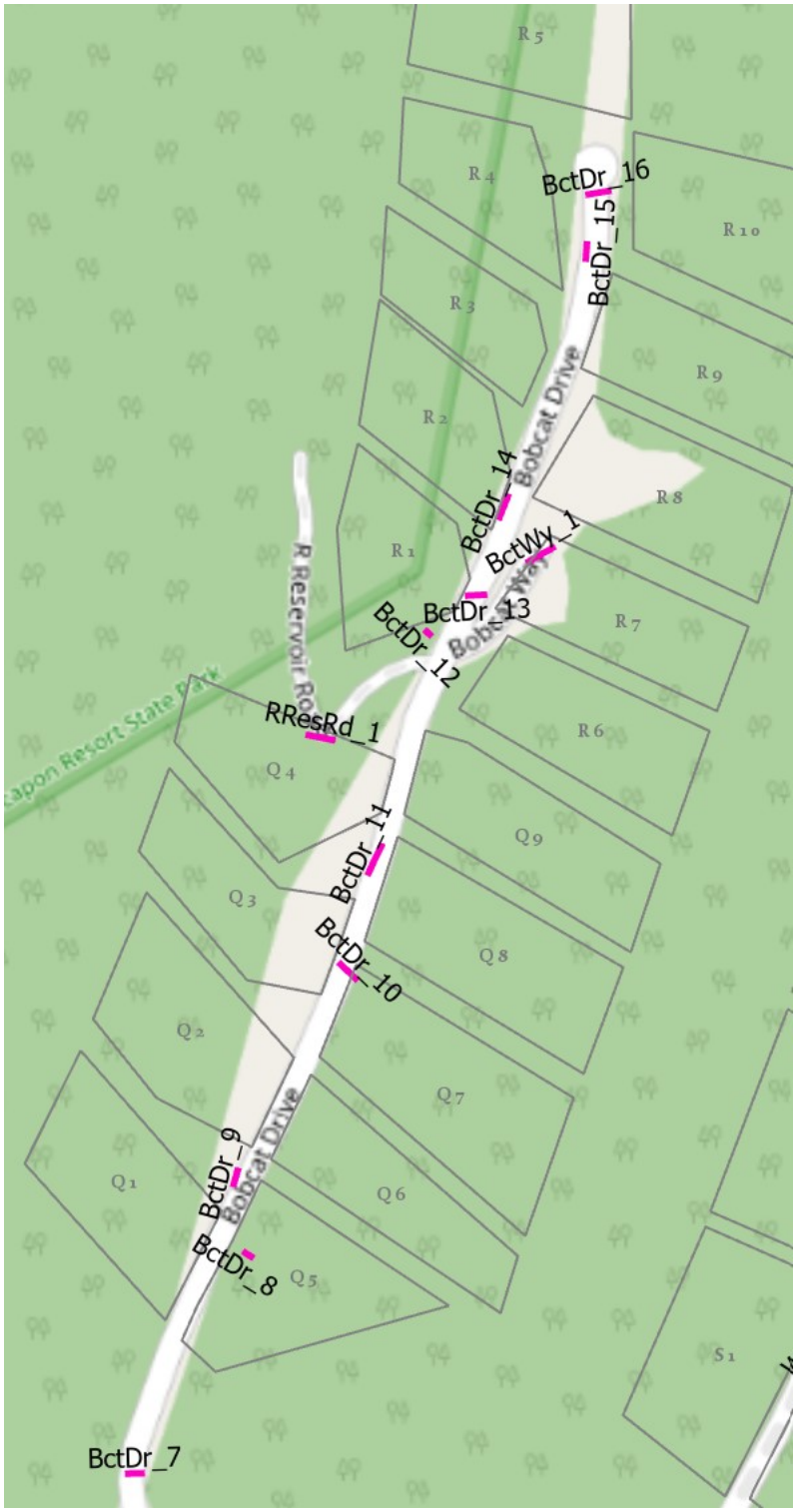
Whippoorwil Way - Mountainside Rd - Frog Hollow Way



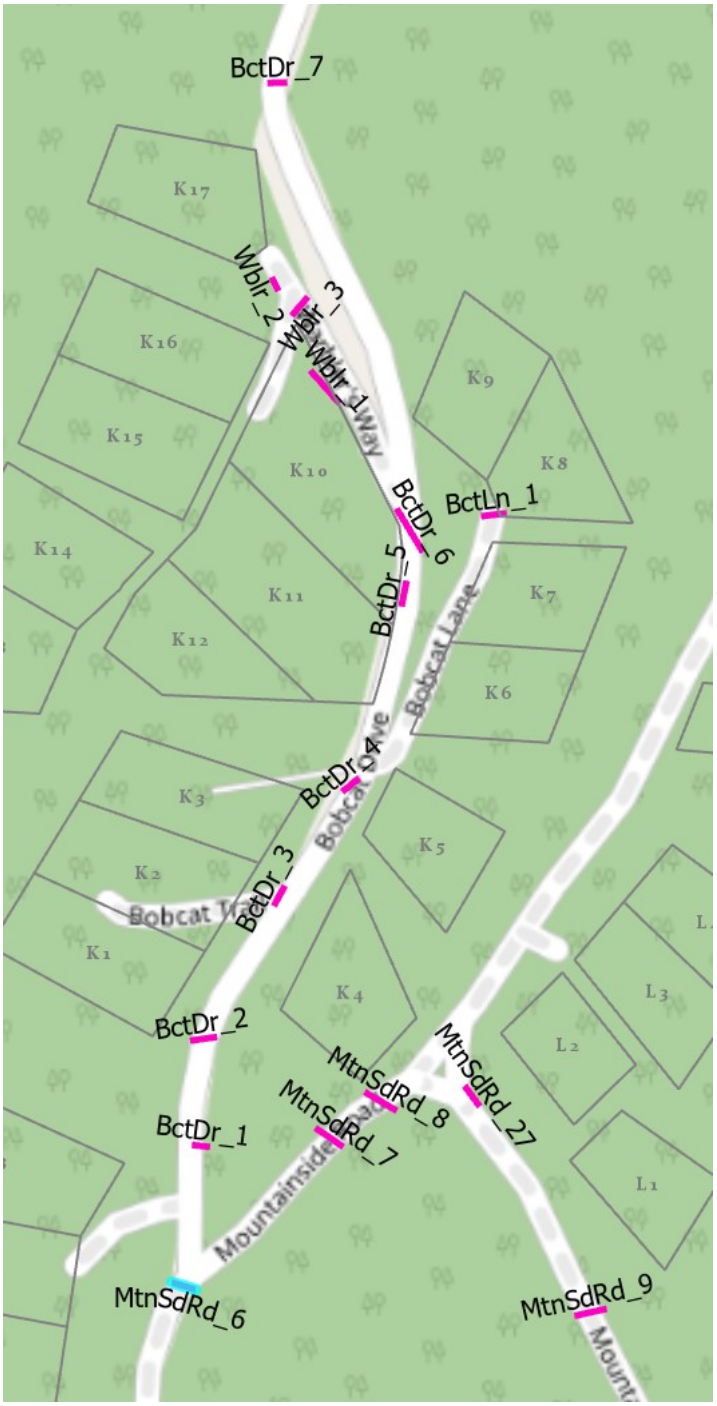
Woodpecker - Mountainside Rd



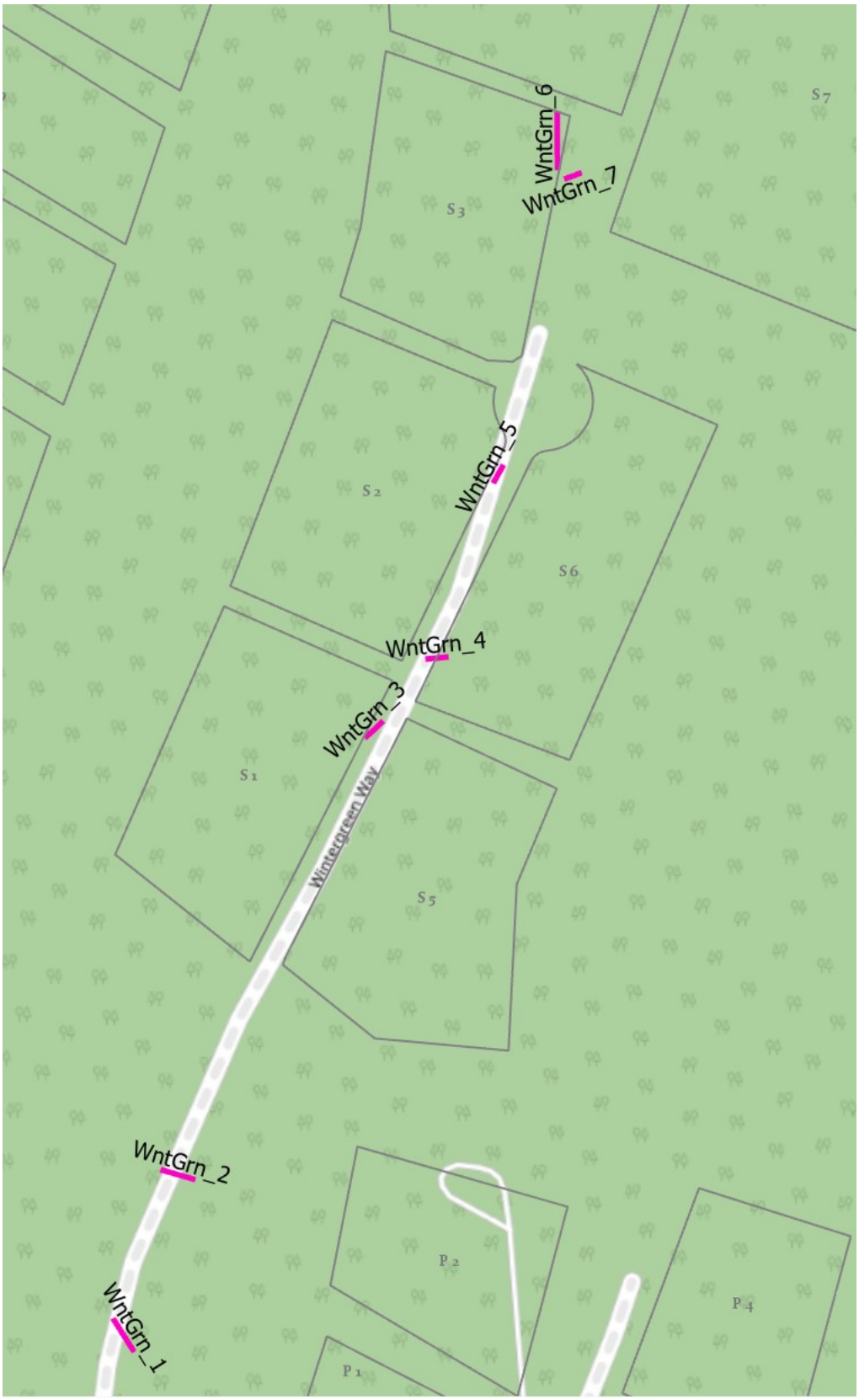
Bobcat Dr North - R Reservoir Access



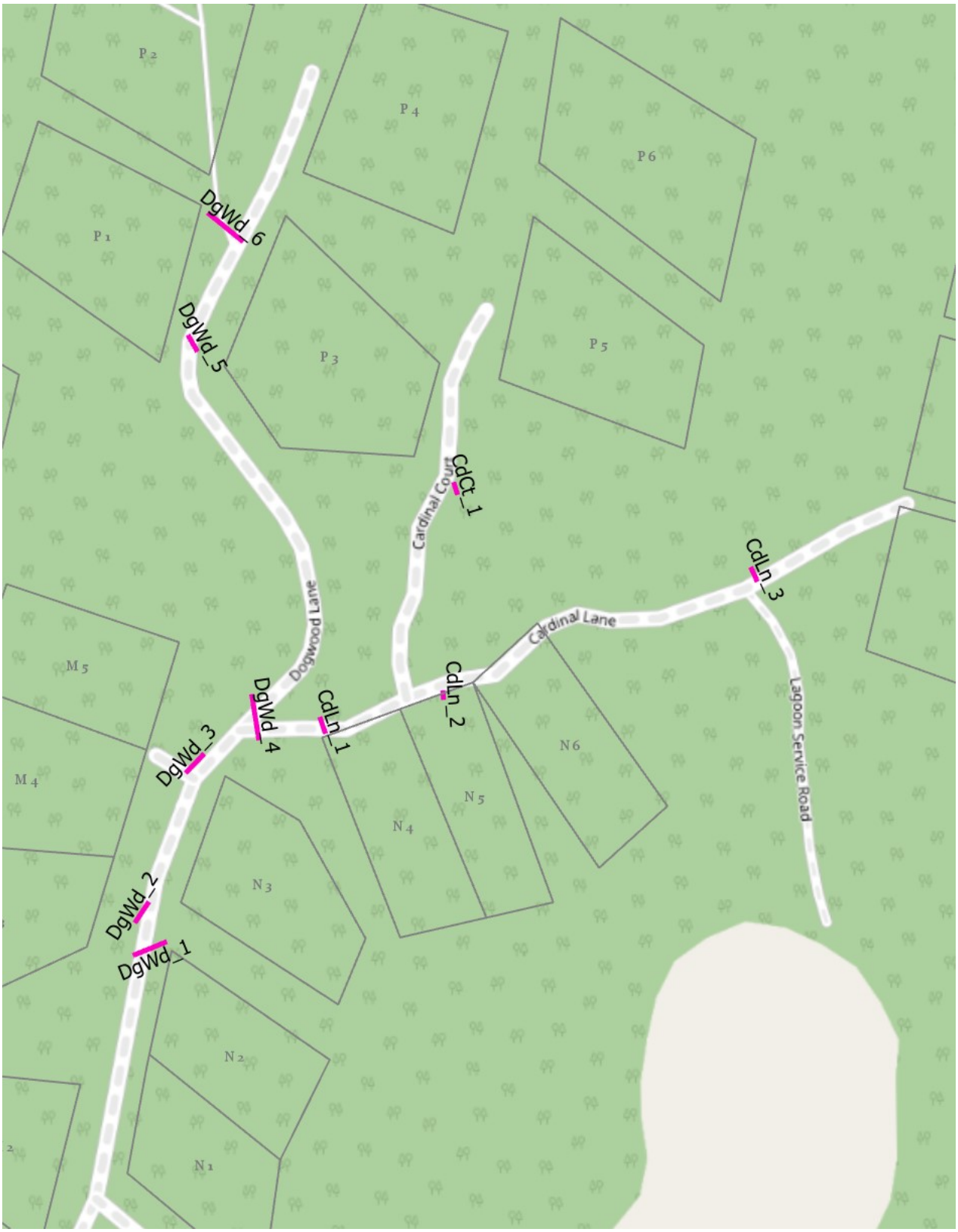
Bobcat Dr South - Bobcat Lane - Mountainside Rd



Wintergreen Way



Dogwood Lane - Cardinal Ln - Cardinal Ct



Mountainside Rd - Herbert Quick

